Task 1

Project Name: UMass Marketplace
Group Name: Group 10

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Task 2

System Boundary:

The application boundary is defined as the UMass Marketplace backend system.
Any data or processes coming from or going to users, external services, or third-party systems are considered outside the application boundary.

Source of Function Counts:

These function point counts were derived from the UMass Marketplace system design, including the project data model, endpoint routes, and use cases such as Posting, Messaging, Filter & Search, and Seller Details.

1. Internal Logical Files (ILF)

- **Users** - stores user account information such as name, email, role, and authentication credentials.
- **Listings** - stores marketplace listings including title, price, status, and associated seller.
- **Products** - stores reusable product information such as name, description, and base details.
- **Categories** - stores predefined categories used to organize listings.
- **Sellers** - stores seller-specific profile information.
- **Conversations** - stores conversation threads between buyers and sellers.
- **Messages** - stores individual messages within conversations.
- **Favorites** - stores user-saved listings for quick access.
- **Transactions** - stores purchase and sale records between buyers and sellers.
- **Reports** - stores reports submitted for moderation purposes.
- **Blocks** - stores blocked user relationships for safety.
- **Photos** - stores image references associated with listings.

These are Internal Logical Files because they are user-identifiable groups of related data that are stored and maintained within our system database. Each of these files represents a persistent data entity that is created, updated, and maintained by the system, and is directly identifiable by users (e.g., listings, users, transactions).

2. External Interface Files (EIF)

None

The system does not rely on external data sources maintained by other applications. All marketplace data is stored and managed internally within the MongoDB database.

3. External Inputs (EI)

- **Register user** - receives user information and creates a new user account in the system.
- **Login user** - receives user credentials and authenticates the user to generate a session/token.
- **Create listing** - receives listing details and inserts a new listing into the database.
- **Update listing** - receives updated listing information and modifies an existing listing.
- **Delete listing** - removes a listing from the system based on user request.
- **Create product** - receives product details and creates a new product record.
- **Update product** - updates existing product information in the system.
- **Delete product** - removes a product record from the database.
- **Create category** - receives category information and adds a new category.
- **Create seller** - creates a seller profile associated with a user.
- **Send message** - receives message content and stores it within a conversation.
- **Create transaction** - records a transaction between buyer and seller.
- **Report listing/user** - submits a report for moderation purposes.
- **Block buyer** - creates a block relationship to restrict interaction between users.
- **Upload photo** - receives image data and stores a reference to the listing.

These are External Inputs because they are unique elementary processes that receive data from outside the application boundary and modify internal system data.

Each External Input represents a user-triggered action that results in a change to internal system data (create, update, or delete operations).

4. External Outputs (EO)

- **Get all listings with filters, pagination, sorting** - returns a processed list of listings based on search queries, category filters, price range, and pagination.
- **Get my listings** - returns listings created by the currently logged-in user.
- Get transaction history - returns processed transaction records associated with a user.
- **Get conversations** - returns conversation threads involving the user, including related metadata.
- **Get favorites** - returns a list of listings saved by the user.

- **Report status results** - returns processed information about submitted reports and their current status.
- **Listing details with populated seller/product/category info** - returns a listing with related data populated from multiple entities such as seller, product, and category.

These are External Outputs because they are unique elementary processes that send processed data outside the application boundary and include derived or computed results.

Each External Output involves additional processing such as filtering, aggregation, or combining data from multiple entities before returning results to the user.

5. External Inquiries (EQ)

- **Get listing by ID** - retrieves a specific listing based on its ID without additional processing.
- **Get product by ID** - retrieves product details based on the product ID.
- **Get seller by ID** - retrieves seller profile information based on seller ID.
- **Get categories** - retrieves the list of available categories.
- **Get current logged-in user** - retrieves the profile of the authenticated user.
- **Get conversation messages** - retrieves messages within a conversation without additional transformation.
- **Get photos for a listing** - retrieves photo references associated with a listing.

These are External Inquiries because they retrieve data from the system and return it to the user without significant processing or transformation.

External Outputs involve additional processing such as filtering, sorting, or data aggregation, while External Inquiries retrieve data without significant transformation.

Each External Inquiry retrieves data directly without performing significant processing, transformation, or business logic.

Function Point Calculation

Type	**Count**	**Complexity**	**Weight**	**Total**
ILF	12	Average	10	120
EIF	0	Average	7	0
EI	15	Average	4	60
EO	7	Average	5	35
EQ	7	Average	4	28

Total FP				**243**

Unadjusted Function Points (UFP) = 243

All components were assumed to have average complexity based on moderate data relationships, validation logic, and typical CRUD operations across the system.

The counts are based on the full UMass Marketplace system, not only the subset implemented by one individual team member.

Overall, the system includes a balanced distribution of data storage (ILFs) and user interactions (EI, EO, EQ), reflecting a moderately complex marketplace application.

This analysis follows the IFPUG Function Point methodology and provides an estimate of the functional size of the UMass Marketplace systems.